# Model Beta B07 Synthetic Sampling Methods Study

HELIX FORGE LAB / p04-full-004031 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000011 (research/programs/model-beta/experiments/results/2026-07-13/model-beta-b07/dataset-assurance/specifications/model-beta-b07-dataset

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 222 | 0.869 | 0.868 | 0.868 |
| slice-02 | 180 | 0.886 | 0.881 | 0.883 |
| slice-03 | 178 | 0.868 | 0.857 | 0.862 |
| slice-04 | 195 | 0.885 | 0.870 | 0.877 |
| slice-05 | 222 | 0.867 | 0.883 | 0.875 |